

The TCP/IP Protocol Stack

Course notes - Department of Computer Science

1. Layered architecture

Networking is organised in layers so each can be designed and replaced independently. The link layer moves frames between directly connected nodes, the network layer routes packets across networks, the transport layer provides process-to-process channels, and the application layer defines what the exchanged data means.

Each layer adds a header on the way down and strips it on the way up. This encapsulation is what allows HTTP to run unchanged over Ethernet, Wi-Fi or a cellular link.

2. Addressing and forwarding

An IPv4 address is 32 bits, conventionally written as four decimal octets, and is split into network and host portions by a prefix length. Classless inter-domain routing allocates prefixes of any length, which slowed address exhaustion and keeps routing tables aggregated.

A router forwards a packet by finding the longest matching prefix in its table and sending the packet to the corresponding next hop. Forwarding is the per-packet operation; routing is the background protocol that populates the table.

3. Reliable transport

IP is best-effort: packets may be dropped, duplicated or reordered. TCP builds a reliable ordered byte stream on top using sequence numbers, cumulative acknowledgements and retransmission on timeout or duplicate acknowledgement.

The connection is established with a three-way handshake that synchronises initial sequence numbers in both directions. UDP omits all of this, offering only multiplexing and an optional checksum, which suits applications that prefer timeliness to completeness.

4. Flow and congestion control

Flow control stops a fast sender from overwhelming a slow receiver, using the advertised window the receiver returns in every acknowledgement.

Congestion control protects the network itself. TCP probes for available capacity with slow start, growing the congestion window exponentially until loss occurs, then switches to additive-increase multiplicative-decrease. Treating loss as a congestion signal is the assumption that keeps the shared network stable.

5. Naming and the application layer

DNS resolves hierarchical names to addresses through a delegated tree of authoritative servers, with aggressive caching at every level to keep the load tractable.

HTTP is a request-response protocol whose statelessness is what makes it scale;

cookies and tokens reintroduce session state where an application needs it. TLS wraps the transport in authenticated encryption, which is now the default rather than the exception.